SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: MLA 0202 **COURSE NAME: Fundamentals of Machine Learning**

COURSE OUTCOME COVERED

CO1: Analyse machine learning fundamentals and mathematical foundations, including probability theory, Bayes decision theory, and information theory. *(BL3)*

Assessment Tool Weightage:40%

QUESTIONS: 10
Marks:50 **Total**

Annexure A

Application-Based Questions

Code of Conduct:

I Kruthika Sre S / 192525360 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Application-Based Questions

Q1 Implement the concept of probability distribution to predict the likelihood of spam emails in a mail filtering system. (5 Mark- 6 Min)

Q2 Illustrate how Bayes Decision Theory can be applied in medical diagnosis for disease prediction. (5 Mark- 6 Min)

Q3 Develop the design steps of a learning system to build a machine learning solution for student performance prediction. (5 Mark- 6 Min)

Q4 Model the outcome of a weather forecasting problem using random variables and probability theory. (5 Mark- 6 Min)

Q5 Measure the uncertainty in a decision-making dataset by applying information theory concepts. (5 Mark- 6 Min)

Q6 Apply conditional probability concepts to design a fraud detection model for suspicious banking transactions. (5 Mark- 6 Min)

Q7 Demonstrate the use of Bayes theorem in predicting customer purchase behavior from prior shopping history. (5 Mark- 6 Min)

Q8 Construct a machine learning workflow to classify handwritten digits by applying the stages of a learning system. (5 Mark- 6 Min)

Q9 Simulate a real-world stock market prediction problem using random variables and suitable probability distributions. (5 Mark- 6 Min)

Q10 Quantify the uncertainty reduction achieved after feature selection using information gain in a classification dataset. (5 Mark- 6 Min)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

37/50

A. probability distribution can be used to find the likelihood that an email is Spam or Not Spam.

1. Collect a dataset of emails labelled as Spam or Not Spam.
2. Identify features such as words like free, offer, win, prize, etc.
3. Calculate the propability of each class:

   P(Spam), P(NotSpam)

4. Assign the email to the class with the higher Probability. Example:

If (Spam | Email) = 0.85, P(NotSpam | Email) = 0.15

Thus, probability distribution helps the mail filtering make predictions based on the likelihood of different classes.

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2) Bayes Decision Theory uses probabilities to choose the most likely disease based on observed sysptoms or test results.

1. Identify possible diseases, such as Disease and No Disease.
2. Calculate the prior Probability of each disease.
3. Obtain medical evidence such as symptoms or test results.
4. Use Baye's Theorem:

$$P(\text{Disease}/\text{Evidence}) = \frac{P(\text{Evidence}\,(\text{Disease})\,P(\text{Disease})}{P(\text{Evidence})}$$

For example, if a patient has certain sysmptoms and:

P(Disease / Symptoms) = 0.80

Thus, Bayes Decision Theory helps doctors make decisions under uncertainty using available medical evidence.

3) The following steps can be used to build a student performance prediction system:

- Define the problem: Predict whether a student will perform well or poorly.
- Collect data: Gather attendance, previous marks, study hours, assignments, etc.
- Prepare data: Remove errors and missing values and convert data into a suitable format.
- Select features: choosen important factors such as attendance and previous marks.
- Train the model: Use a machine learning algorithm such as Decision Tree or Logistic Regression.
- Test and evaluate: Test the model using unseen student data and measure accuracy.

4) Weather conditions can be represented using random variables because the exact future weather is uncertain.

For example, let: X = Tomorrow's weather

Possible outcomes are:

- X = Sunny
- X = Rainy
- X = Cloudy

Assign possibilities such as:

$P(\text{sunny}) = 0.5$

$P(\text{Rainy}) = 0.3$

$P(\text{cloudy}) = 0.2$

The probabilities add up to 1 : $0.5 + 0.3 + 0.2 = 1$

5) Information theory of a dataset is calculated as:

$$H(S) = -\sum p_i \log_2(p_i)$$

For example, suppose a dataset has:

- 6 Yes decisions
- 4 No decisions

Then: $H(S) = -(0.6 \log_2 0.6 + 0.4 \log_2 0.4) \approx 0.971$

A higher entropy means greater uncertainty, while lower entropy means the data is more certain.

6) Conditional probability can determine the probability that a transaction is fraudulent given certain suspicious features.

Let:
- F = transaction is fraudulent
- E = suspicious evidence, such as unusual location or high amount.

Calculate: $$P(F|E) = \frac{P(E|F)\,P(F)}{P(E)}$$

Steps:
1. Collect previous banking transactions labelled as fradulent or genuine.
2. Identify features such as transaction amount, location and frequency.
3. Calculate conditional probabilities from historical data.
4. For a new transaction, calculate $P(F|E)$.
5. If the probability is above a chosen threshold, flag the transaction as fradulent/suspicious.

7) Bayes theorem can predict whether a customer is likely to purchase a product based on previous shopping behaviour.

Formula:

$$P(\text{Purchase}|\text{History}) = \frac{P(\text{History}|\text{Purchase})\,P(\text{Purchase})}{P(\text{History})}$$

Steps:
1. Collect customer's previous purchase history.
2. Calculate the prior probability of purchasing a product.
3. Observe the customer's previous shopping behaviour.
4. Calculate the posterior probability of purchase.
5. If the probability is high, recommend the product to the customer.

For example, if:

$$P(\text{Purchase}|\text{History}) = 0.75$$

8) Collect data: Use labelled handwritten digit images such as the MNIST dataset.
Preprocess data: Convert images into suitable numerical pixel values and normalize them.
Extract/select features: Use pixel patterns that help distinguish digits
Train the model: Apply an algorithm such as kNN, Decision Tree or Neural Network
Test the model: Given unseen handwritten digit images to the trained model.
Evaluate: Measure accuracy by comparing predicted digits with actual digits.

9) Stocks prices are uncertain, so they can be represented using random variables. Let: X = daily stock return.
Historical stock data can be used to estimate the mean and standard deviation of returns. A suitable distribution, such as the Normal distribution, can be used to model the returns: $X \sim N(\mu, \sigma^2)$

Steps:
1. Collect historial stock prices.
2. Calculate daily returns.
3. Estimate the mean μ and standard deviation σ.
4. Generate random returns using the selected probability distribution.
5. Use these simulated values to estimate possible future stock prices.

10) Information Gain measures how much uncertainty is reduced when a feature is used for classification.
It is calculated as:

$$IG(S, A) = H(S) - H(S/A)$$

where,
- $H(S)$ = entropy before selecting the feature
- $H(S/A)$ = entropy after using feature A

Example: if the original entropy is:

$$H(S) = 0.97$$

and after selecting a feature:

$$H(S/A) = 0.40$$

then: $IG = 0.97 - 0.40 = 0.57$
∴ The feature reduces uncertainty by 0.57 bits.